Methods note: how the six baselines are calculated

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Documentation only: this note describes the code as frozen (git tag `pre-test-freeze`); no method, parameter or result is changed. Function names refer to `src/sst/`.

1. Notation

- **s(t, i)**: daily OISST v2.1 sea-surface temperature (°C) on day *t* at grid point *i* (0.25° grid; two 40×40 regions, Kuroshio 30.125–39.875°N, 145.125–154.875°E and subtropical 15.125–24.875°N, 150.125–159.875°E). Both boxes are open ocean (no NaN points; the code would skip NaN points).
- **t** is the issue date (the last day whose SST is known), **h** the lead (1, 7, 30 days), **t+h** the target date. A forecast issued at *t* uses SST only on days ≤ *t*.
- **d(t)** ∈ {1,...,365} is the calendar day in a 365-day calendar: 1 Jan = 1, 31 Dec = 365, and 29 Feb is mapped to 28 Feb (`climatology.doy365`).
- Periods: training 1982–2015, validation 2016–2019, test 2020–2025. Everything fitted (climatology, *r*(h), Ridge) uses training years only.
- Every baseline is computed separately at each grid point; the subscript *i* is dropped below when it is not needed.

2. Climatology and anomaly

Raw day-of-year mean (`climatology.raw_doy_mean`): for each calendar day *d*, the average of s over all training days with d(t) = *d* (34 years; 29 Feb observations are pooled into 28 Feb, so 28 Feb averages 42 values (34 + 8 leap days), other days 34; the regions have no missing days, and the code skips them if present):

$c_{raw}(d, i) = \text{mean}\{\ s(t, i) : t \text{ in } 1982–2015,\ d(t) = d\ \}.$

Smoothed climatology (`climatology.smooth_harmonics`, `build_climatology`, config `smoothing: harmonics3`): at each grid point, c(·, i) is the least-squares fit (`numpy.linalg.lstsq`) of the 365 raw values by a constant and three annual harmonics,

$$c(d, i) = \beta_0 + \Sigma_{k=1..3}\ [\ \beta_{k,c}\cos(2\pi k(d-1)/365) + \beta_{k,s}\sin(2\pi k(d-1)/365)\ ].$$

Seven coefficients per grid point, fitted on the training raw means only (equal weights; the angle uses d−1, i.e. the 0-based index used in the code). For the grid point of §5 (35.125°N, 147.125°E): β_0 = 20.9395, ($\beta_{1,c}$, $\beta_{1,s}$) = (-2.4666, -4.4852), ($\beta_{2,c}$, $\beta_{2,s}$) = (-0.1940, 0.7230), ($\beta_{3,c}$, $\beta_{3,s}$) = (0.2891, -0.1761) °C; the RMS difference between raw and fitted values over the 365 days is 0.166 °C. The raw mean is only an intermediate; every forecast and anomaly uses the smoothed c. `clim_at` looks c up for any date (any year, including validation and test) through d(t).

Anomaly (`climatology.anomaly`): a(t, i) = s(t, i) − c(d(t), i). The climatology is never refitted on later years, so a warm shift after 2015 stays in the anomaly.

3. The six baselines

Forecasts for target *t*+*h*, written ŝ(t+h). All are evaluated on the same (issue, target) pairs (`metrics.valid_pairs`; a pair is skipped and counted if the issue or target day is missing; `run_val_all.run_region` asserts that Ridge uses identical pairs).

(1) Persistence (`baselines.persistence`): ŝ(t+h) = s(t). No climatology, no parameters.

(2) Fixed smoothed climatology (`baselines.climatology_forecast`; called "Climatology" in tables and figures): ŝ(t+h) = c(d(t+h)), the 1982–2015 3-harmonic fit of §2. It ignores s(t) and gives the same value for every year and every lead at a given target day.

(3) Anomaly persistence (`baselines.anomaly_persistence`): ŝ(t+h) = c(d(t+h)) + a(t). Today's anomaly is kept unchanged (equivalent to s(t) + [c(d(t+h)) − c(d(t))]).

(4) Damped persistence (`baselines.lag_corr`, `damped_persistence`): ŝ(t+h) = c(d(t+h)) + $r_i(h)$·a(t). One coefficient per grid point and lead:

$$r_i(h) = \Sigma\ a(t,i)\cdot a(t+h,i)\ /\ \sqrt{(\ \Sigma\ a(t,i)^2 \cdot \Sigma\ a(t+h,i)^2\ )},$$

summed over all pairs (t, t+h) in which **both** days lie in the training years 1982–2015 (all seasons pooled; for this example 12,411 pairs). It is **not centred**: no mean is subtracted (the training-anomaly means are ≈ 0, e.g. +0.0003 °C here, so it is almost the Pearson correlation). Validation and test data are never used; the same $r_i(h)$ is applied in every later year. It is a correlation, not the regression slope $\Sigma\ a(t)a(t+h)\ /\ \Sigma\ a(t)^2$; the two are close because the variances of a(t) and a(t+h) are nearly equal (here 18273 vs 18194).

(5) 10-year daily climatology, "OCN" (`baselines.ocn`, `ocn_years: 10`): for target date T = t+h,

$$\hat{s}(t+h) = (1/10)\ \Sigma_{k=1..10}\ s(T - k\ \text{years}),$$

i.e. the mean of the **same calendar day as the target** in the 10 years before the **target** year (T minus 1, 2, ..., 10 years; the window is anchored on the target, not on the issue date or the end of training). If T is 29 Feb, 28 Feb is used first and then shifted back, so 29 Feb of earlier leap years is never read. The single-day values are averaged as they are: no smoothing, no anomalies, no climatology, no training split; missing days are skipped by `nanmean`. All ten dates lie at least one year before T; the code raises a `ValueError` unless every date is strictly before the issue date (true for h < 365 d), so the baseline uses only information available at *t*. The window moves with the target (e.g. for 2021 targets it covers 2011–2020), so it can contain validation/test years that precede the issue date.

(6) Ridge (`ridge.make_features`, `fit_ridge`, `predict_ridge`, `ridge_lead`): one linear model per grid point and per lead.

- Features at issue date t (8): the anomalies a(t), a(t−1), ..., a(t−6) (lags 0–6) and the mean anomaly $\bar{a}_{30}(t) = (1/30)\ \Sigma_{j=0..29}\ a(t-j)$ over t−29..t (anomalies relative to the training climatology). A pair needs all 30 days present.
- Target: the anomaly y = a(t+h) = s(t+h) − c(d(t+h)).
- Training sample: all issue dates whose target lies in 1982–2015 and whose 30-day history is complete (`_pairs`; 12382 rows for h = 7 at the §5 point). Features read only days ≤ t.
- Standardisation: each feature is z = (x − μ)/σ, where μ and σ (population std) are the mean and standard deviation of that feature over the training rows of that grid point and lead; the target is centred by its training

mean $\bar{y}$ (the intercept, not penalised).
- Fit: $w = \text{argmin}\ (1/n)\ \Sigma\ (y - \bar{y} - z^T w)^2 + \alpha\|w\|^2$, solved in closed form as $w = (Z^T Z/n + \alpha I)^{-1}\ Z^T (y - \bar{y})/n$.
- α: grid 10^{-3}, $10^{-2.5}$, ..., 10^{3} (13 values). For each region and lead, one α (shared by all grid points of the region) is chosen as the value with the lowest cos(lat)-weighted RMSE of the forecast SST on validation 2016–2019 (`ridge_lead`, validation branch). Chosen values (config.yaml, `results/ridge_alphas.json`): Kuroshio 0.00316228 / 0.001 / 0.001, subtropical 0.01 / 0.001 / 0.001 for h = 1 / 7 / 30 d; several lie at the lower edge of the grid. The final model is fitted on **training years only** with this α (no refit on validation) and applied unchanged to the test period.
- Forecast: $\hat{s}(t+h) = c(d(t+h)) + \hat{y}$, with $\hat{y} = \bar{y} + \Sigma_{m=1..8}\ w_m\ z_m$.

4. Scores

With weights $w_j = \cos \varphi_j\ /\ \text{mean}_j(\cos \varphi_j)$ for latitude φ_j (`metrics.lat_weights`), $e = \hat{s} - s$ over all evaluated pairs and valid grid points:

- **RMSE** = $\sqrt{(\ \Sigma\ w \cdot e^2\ /\ \Sigma\ w\ )}$; **MAE** = $\Sigma\ w \cdot |e|\ /\ \Sigma\ w$ (`rmse`, `mae`; one pooled number per region, lead and method).
- **ACC** (temporal, primary; `metrics.acc`): with forecast anomaly $f' = \hat{s} - c(d(t+h))$ and observed anomaly $o' = s(t+h) - c(d(t+h))$, each centred by its own mean over the evaluated target dates at that grid point, $r_i = \Sigma_t\ f' o'\ /\ \sqrt{(\Sigma\ f'^2\ \Sigma\ o'^2)}$; then ACC = $\Sigma_i\ w_i\ r_i\ /\ \Sigma_i\ w_i$ over points where r_i is defined (zero-variance points, e.g. the climatology forecast, are skipped, so ACC is undefined for it). A secondary daily spatial ACC (`acc_spatial`) is also stored.
- **Skill** (`metrics.skill_score`) = $1 - \text{RMSE}_{\text{model}}\ /\ \text{RMSE}_{\text{ref}}$; primary reference anomaly persistence, also computed against the fixed climatology (0 = no better than the reference, 1 = perfect, negative = worse).

5. Worked example (real data, lead h = 7 d)

Grid point 35.125°N, 147.125°E (Kuroshio region). Issue date t = 2021-03-15 (test period), target t+h = 2021-03-22. Values are computed at full precision and displayed to 3 decimals; errors are forecast − observed. Nothing is refitted: the climatology, r(h) and Ridge model are the training-year objects used in the test run.

Quantity	Value
s(t), 2021-03-15	17.160 °C
d(t), d(t+h)	74, 81
c(d(t)) (raw mean 16.423)	16.363 °C
c(d(t+h)) (raw mean 16.366)	16.505 °C
a(t) = s(t) − c(d(t))	+0.797 °C
r(h) = 11878.2 / √(18273.4 · 18193.8), 12,411 training pairs	0.6514
Observed s(t+h), 2021-03-22	**14.360 °C**

OCN (same calendar day as the target, 22 March, 10 years before 2021; latest date 2020-03-22, 358 days before the issue date):

Year	2020	2019	2018	2017	2016	2015	2014	2013	2012	2011	Mean
s (°C)	17.18	16.68	17.09	17.01	18.43	16.66	14.77	14.87	17.09	15.41	**16.519**

Ridge (α = 0.001, intercept $\bar{y}$ = 0.0013 °C):

Feature	a(t)	a(t-1)	a(t-2)	a(t-3)	a(t-4)	a(t-5)	a(t-6)	mean a(t-29..t)
x (°C)	0.797	0.903	0.607	-0.089	0.533	0.714	0.424	0.387
μ (train)	0.0019	0.0019	0.0020	0.0021	0.0023	0.0024	0.0026	0.0025
σ (train)	1.212	1.212	1.212	1.212	1.212	1.212	1.212	0.960
z = (x−μ)/σ	0.656	0.743	0.500	-0.075	0.438	0.587	0.348	0.400
w	0.723	-0.204	0.102	0.030	0.022	0.020	0.005	0.166
w·z	0.474	-0.152	0.051	-0.002	0.009	0.012	0.002	0.066

$\hat{y} = \bar{y} + \Sigma\ w \cdot z = 0.0013 + 0.4603 = +0.462$ °C.

The six forecasts (target observed 14.360 °C):

Baseline	Formula with numbers	Forecast (°C)	Error (°C)
Persistence	s(t)	17.160	+2.800
Fixed smoothed climatology	c(d(t+h)) = 16.505	16.505	+2.145
Anomaly persistence	16.505 + 0.797	17.302	+2.942
Damped persistence	16.505 + 0.6514 · 0.797	17.024	+2.664
OCN (10-yr daily climatology)	mean of the 10 values above	16.519	+2.159
Ridge	16.505 + 0.4616	16.966	+2.606

On this date SST fell by 2.80 °C within the week (a cold event), so every method overshoots; the methods that ignore today's warm anomaly (fixed climatology, OCN) happen to be closest. This is a single illustrative case, not a measure of skill. Reproduce: `python report/methods_example.py` recomputes each step by hand-written arithmetic, asserts agreement with the project functions (and with the full-region `ridge_lead` output) and writes `results/methods_example.json`; this note is generated from that file by `report/build_methods_note.py`.

6. Naming note

In oceanography a "daily climatology" normally means the mean of each calendar day over a set of years. In this project, **OCN** is such a daily climatology computed over a **moving 10-year window** ending the year before the target (raw, unsmoothed, anchored on the target date), whereas the **fixed smoothed climatology** (called "Climatology" in the tables, figures and README) is the 3-harmonic fit to the 1982–2015 day-of-year means. The two differ in window (moving 10 years vs fixed 34 training years) and in smoothing (none vs three harmonics). This note uses these two names consistently; no names have been changed in the code or README.